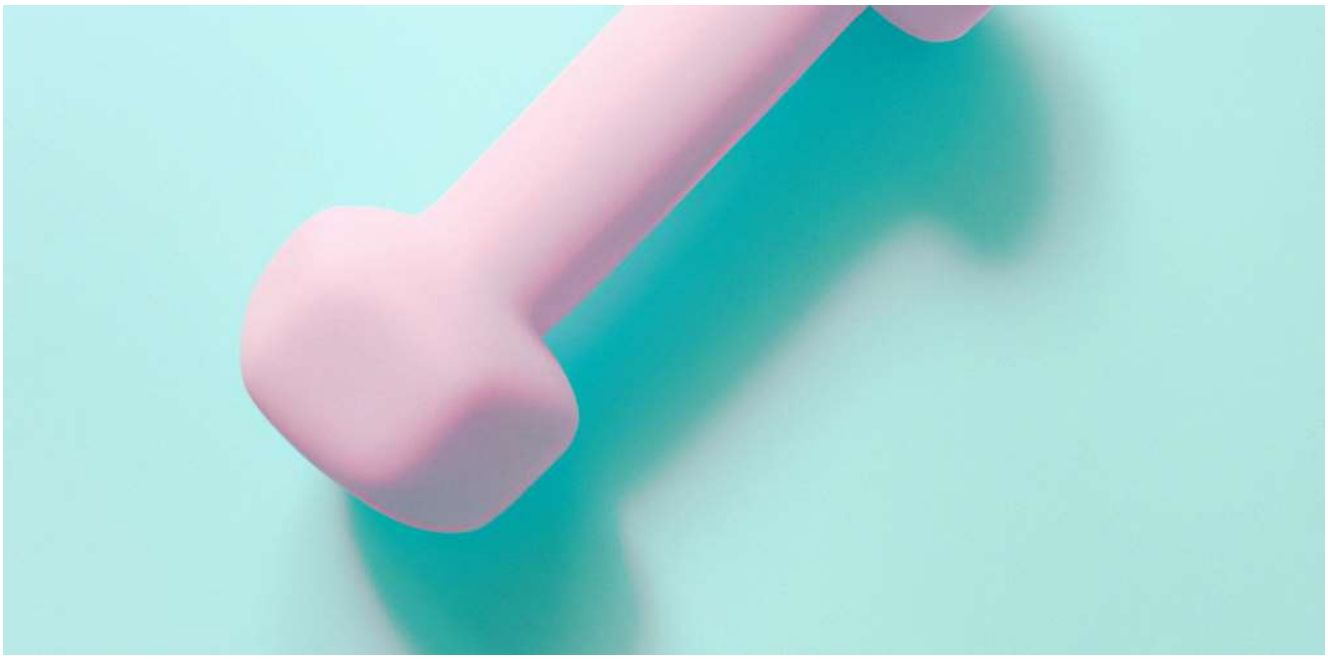
PROJECT: DATA-DRIVEN PRODUCT MANAGEMENT: CONDUCTING A MARKET ANALY…



You are a product manager for a fitness studio and are interested in understanding the current demand for digital fitness classes. You plan to conduct a market analysis in Python to gauge demand and identify potential areas for growth of digital products and services.

The Data

You are provided with a number of CSV files in the "Files/data" folder, which offer international and national-level data on Google Trends keyword searches related to fitness and related products.

workout.csv

Column	Description
`'month'`	Month when the data was measured.
`'workout_worldwide'`	Index representing the popularity of the keyword 'workout', on a scale of 0 to 100.

three_keywords.csv

Column	Description
`'month'`	Month when the data was measured.
`'home_workout_worldwide'`	Index representing the popularity of the keyword 'home workout', on a scale of 0 to 100.
`'gym_workout_worldwide'`	Index representing the popularity of the keyword 'gym workout', on a scale of 0 to 100.
`'home_gym_worldwide'`	Index representing the popularity of the keyword 'home gym', on a scale of 0 to 100.

workout_geo.csv

Column	Description
`'country'`	Country where the data was measured.
`'workout_2018_2023'`	Index representing the popularity of the keyword 'workout' during the 5 year period.

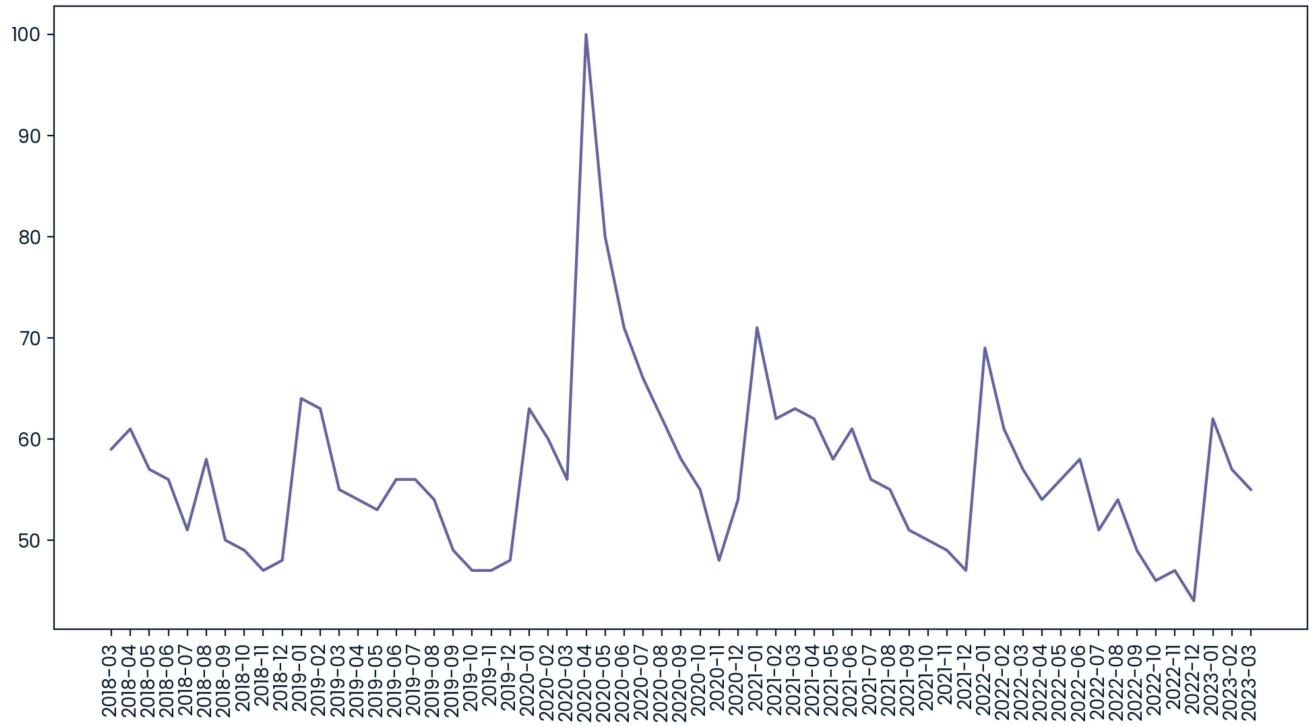
three_keywords_geo.csv

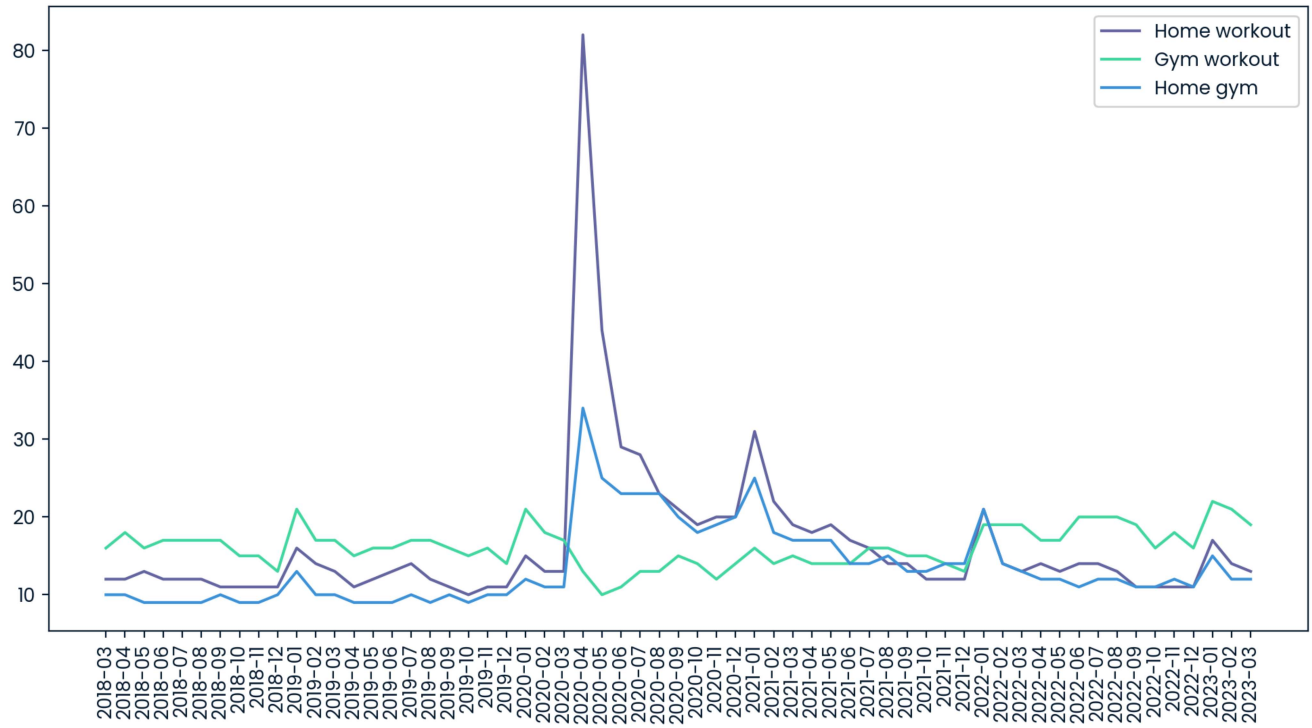
Column	Description
`'country'`	Country where the data was measured.

Column	Description
`'home_workout_2018_2023'`	Index representing the popularity of the keyword 'home workout' during the 5 year period.
`'gym_workout_2018_2023'`	Index representing the popularity of the keyword 'gym workout' during the 5 year period.
`'home_gym_2018_2023'`	Index representing the popularity of the keyword 'home gym' during the 5 year period.

```
# Import the necessary libraries
import pandas as pd
import matplotlib.pyplot as plt
```

```
# Start coding here
# Find the peak for global 'workout' searches
df_workout = pd.read_csv("data/workout.csv")

plt.figure(figsize=(12, 6))
plt.plot(df_workout["month"], df_workout["workout_worldwide"])
plt.xticks(rotation=90)
plt.show()

year_str = "2020"

# Find the most popular keywords for the current year and during covid
df_keywords = pd.read_csv("data/three_keywords.csv")

plt.figure(figsize=(12, 6))
plt.plot(df_keywords["month"], df_keywords["home_workout_worldwide"], label="Home
workout")
plt.plot(df_keywords["month"], df_keywords["gym_workout_worldwide"], label="Gym
workout")
plt.plot(df_keywords["month"], df_keywords["home_gym_worldwide"], label="Home gym")
plt.xticks(rotation=90)
plt.legend()
plt.show()

peak_covid = "home workout"
current = "gym workout"

# Find the country with the highest interest for workouts
df_workout_geo = pd.read_csv("data/workout_geo.csv", index_col = 0)
print(df_workout_geo.loc["United States"])
print(df_workout_geo.loc["Australia"])
print(df_workout_geo.loc["Japan"])

top_country = "United States"

# Who has the highest interest in home workouts, Philippines or Malaysia?
df_keywords_geo = pd.read_csv("data/three_keywords_geo.csv", index_col = 0)
print(df_keywords_geo.loc["Philippines", :])
print(df_keywords_geo.loc["Malaysia", :])

home_workout_geo = "Philippines"
```






```
workout_2018_2023    100.0
Name: United States, dtype: float64
workout_2018_2023    77.0
Name: Australia, dtype: float64
workout_2018_2023    1.0
Name: Japan, dtype: float64
home_workout_2018_2023    52.0
gym_workout_2018_2023     38.0
home_gym_2018_2023        10.0
```

```
Name: Philippines, dtype: float64
home_workout_2018_2023    47.0
gym_workout_2018_2023     38.0
home_gym_2018_2023        15.0
Name: Malaysia, dtype: float64
```